Customer Shopping Behavior Analysis

A Python, SQL Server, and Power BI portfolio project

Analysis of 3,900 records describing customer demographics, purchase amounts, product categories, discounts, shipping preferences, subscription status, and review ratings.

Purpose
Turn the supplied shopping dataset into clear, reproducible descriptive analysis and business questions that can be explored in SQL and presented in a Power BI dashboard.

Tools used
Python in Jupyter Notebook; Microsoft SQL Server queried through SQL Server Management Studio (SSMS) 22; and Microsoft Power BI Desktop for the dashboard/report.

Prepared from the supplied CSV, Python notebook, SQL query file, and Power BI report file. A genuine exported dashboard screenshot was not supplied, so a placeholder is retained rather than inventing visual evidence.

1. Executive Summary

The supplied dataset contains 3,900 records. Purchase amounts total $233,081, with an average purchase amount of $59.76 and a median of $60.00. The source file has 37 missing review ratings; the supplied notebook checks missing values but does not fill them. Rating summaries therefore use available values only.

This is descriptive analysis, not proof of causal effects. “Revenue” in this report means the sum of the dataset’s purchase-amount field. Profit, customer lifetime value, and incremental impact cannot be calculated from the provided fields.

2. Business Objective and Questions

3.
The project explores how a retailer could use shopping records to understand purchase patterns and support decisions about products, promotions, subscriptions, shipping, and customer engagement.

The supplied SQL script contains ten analysis questions:

1. Compare purchase-amount totals by gender.
2. Find discount users whose purchase amount is at least the overall average.
3. Rank products by average review rating.

4. Compare average purchase amounts for Standard and Express shipping.
5. Compare record count, average purchase amount, and total purchase amount by subscription status.
6. Find products with the highest share of discounted records.
7. Group records as New, Returning, or Loyal using previous-purchase counts.
8. Rank the three most frequently recorded products within each category.
9. Compare subscription-status counts among records with more than five previous purchases.
10. Compare purchase-amount totals across age groups.

4. Dataset Profile

5.

Attribute	Observed value
Rows / columns	3,900 rows / 18 columns in the supplied raw CSV; the prepared SQL table may contain derived columns.
Customer ID	3,900 distinct IDs; IDs range from 1 to 3900.
Age	18–70 years; mean approximately 44.1.
Purchase amount	$20–$100; mean $59.76.
Categories	Clothing (1,737); Accessories (1,240); Footwear (599); Outerwear (324).
Missing values	37 nulls in review_rating; no nulls in the other supplied raw columns.
Discount vs promo code	The two fields match for all supplied records.

Interpretation note: The file has one row per Customer ID and the SQL pipeline treats each row as an analysis record. The dataset does not include dated transaction history, units, cost, margin, or an explicit online/offline channel field; avoid describing it as a time-series, profitability, or channel-comparison dataset.

4. Python Preparation and Data Quality

The supplied notebook, **customers analysis.ipynb**, uses Python in Jupyter Notebook. It reads the CSV with pandas; inspects rows, schema, descriptive statistics, and null counts; standardizes column names to lowercase snake_case; renames the purchase amount field; creates age_group with quartile-based `pandas.qcut` labels; maps purchase-frequency labels to approximate day counts; checks whether discount_applied and promo_code_used match; drops promo_code_used; and loads the DataFrame into SQL Server using SQLAlchemy and pyodbc.

Important reproducibility notes

• The notebook does not impute the 37 missing review_rating values.

- The notebook is configured for Microsoft SQL Server (localhost, database customer_behavior, WindowsAuthentication, ODBC Driver 17), not PostgreSQL.
- Frequency-to-days values are approximations, not observed dates between purchases. “Fortnightly” and “Bi-Weekly”map to 14 days; “Quarterly” and “Every 3 Months” map to 90 days.
- Since discount_applied and promo_code_used match for all supplied rows, dropping promo_code_used follows thenotebook’s check for this dataset. Recheck if the source changes.
- The notebook uses `if_exists="replace"` when writing the customer table; this replaces the existing table each time the cell runs. Use only when replacement is intended.

5. SQL Analysis Results

The following figures are calculated from the supplied CSV using the filters and grouping logic represented in the supplied SQL questions. Execute the SQL in Microsoft SQL Server using SSMS 22 against the loaded table to verify the same outputs.

5.1 Purchase amount by gender

Gender	Records	Total purchase amount	Average purchase amount
Female	1,248	$75,191	$60.25
Male	2,652	$157,890	$59.54

The male group has a larger total in this dataset alongside a much larger record count. Average purchase amounts are close; do not interpret the total difference as evidence that gender causes spending differences.

5.2 Discount users spending at least the overall average

839 records satisfy discount_applied = 'Yes' and purchase_amount ≥ $59.76. The SQL question returns customer IDs and purchase amounts; it does not measure whether discounts increased spending.

5.3 Top five products by average review rating

Product	Mean available review rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

The SQL does not impute missing ratings; SQL AVG excludes NULL values. Ratings are descriptive and do not by themselves establish product quality or sales performance.

5.4 Standard vs Express shipping

Shipping type	Records	Average purchase amount	Total purchase amount
Express	646	$60.48	$39,067
Standard	654	$58.46	$38,233

This comparison covers only Standard and Express, as specified by the SQL. Other shipping types are excluded.

5.5 Subscribers vs non-subscribers

Subscription status	Records	Average purchase amount	Total purchase amount
No	2,847	$59.87	$170,436
Yes	1,053	$59.49	$62,645

The subscriber average is slightly below the non-subscriber average in this dataset. Subscription status alone is not proof of subscription impact; group sizes differ and the data are observational.

6. SQL Analysis Results (continued)

5.6 Products with the highest discount rates

Product	Records with discount / all product records
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

These rates describe how frequently the discount flag appears for each item. The SQL returns the five highest rates but does not apply a minimum sample-size threshold or measure profitability.

5.7 Customer segmentation by previous purchases

Segment rule in supplied SQL	Records
New (previous_purchases = 1)	83
Returning (2–10)	701
Loyal (all other values)	3,116

This is the exact rule in the supplied SQL. It labels one previous purchase as New, which may be counterintuitive, and labels every value outside 1–10 as Loyal. Review and document the business definition before using these segments.

5.8 Top products within each category

The supplied query counts records for each category/item combination and ranks items within each category using ROW_NUMBER(). It returns up to three products per category. No product ranking results are asserted here because the query does not include a tie-breaker; execute it and capture the real output.

5.9 Repeat buyers and subscription status

Subscription status	Records with previous_purchases > 5
No	2,518
Yes	958

The supplied query returns counts among repeat-buyer records, not subscription rates. To compare likelihood fairly, calculate the subscription proportion within the >5-purchase group and compare it with the proportion among other records.

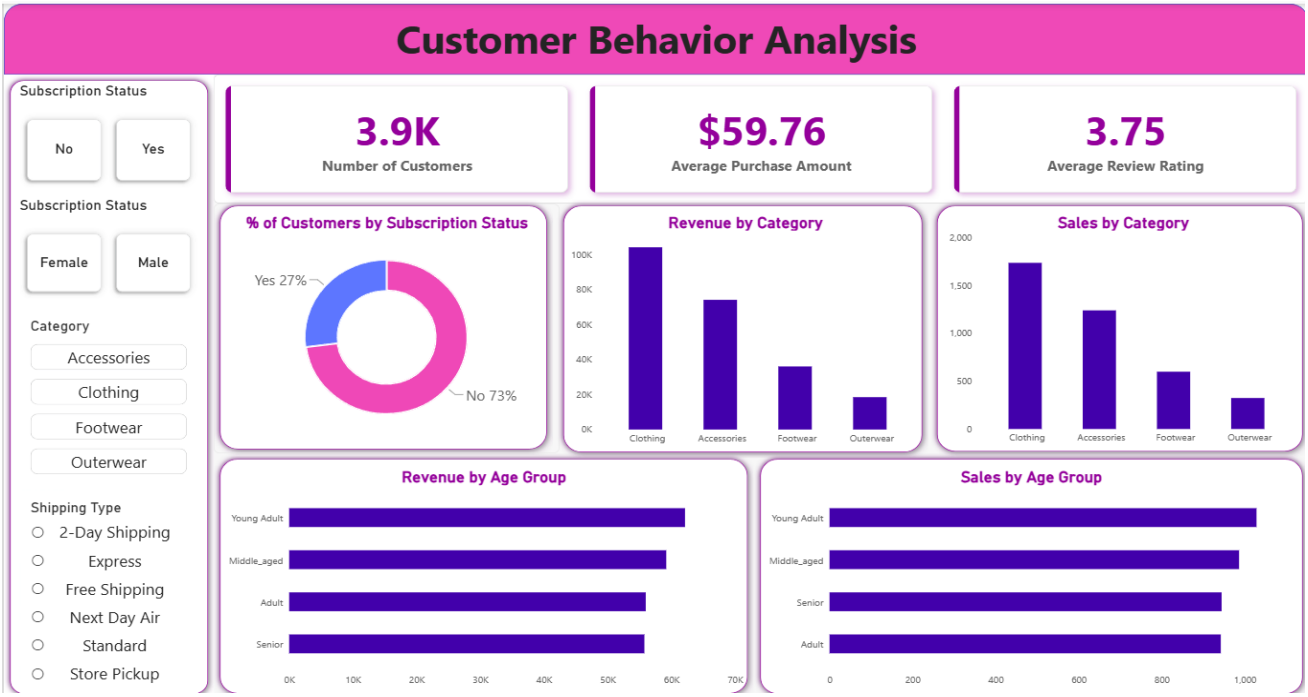
5.10 Purchase amount by age group

Age group	Records	Total purchase amount	Average purchase amount
Young Adult	1,028	$62,143	$60.45
Adult	942	$55,978	$59.42
Middle-aged	986	$59,197	$60.04
Senior	944	$55,763	$59.07

Age groups are created with pandas.qcut, so labels represent data-dependent quartile groups rather than fixed demographic standards. Document the actual boundaries from the executed notebook if they are shown in the dashboard.

7. Power BI Dashboard

The supplied .pbix file is the dashboard/report artifact. Open it in Microsoft Power BI Desktop, check that visuals load and data connections are valid, and compare key totals against the CSV and SQL results. A verified rendered dashboard screenshot was not supplied for this report.



7. Business Recommendations

The dataset supports exploratory recommendations that should be tested before being treated as business decisions:

- **Evaluate subscription value:** compare repeat rate, spend distribution, and retention for subscribers versus non-subscribers. The current average purchase amount alone does not show subscriber uplift.

- **Audit discount effectiveness:** assess conversion, incremental revenue, and margin before increasing discounts.

Discount frequency is not the same as discount profitability.

- **Use product ratings carefully:** show rating averages with the number of non-null ratings and avoid overinterpreting small differences.

- **Plan product-specific merchandising:** combine purchase counts, ratings, and discount rates to find products that are popular, well-rated, or promotion-dependent.

- **Validate customer segments:** agree on business definitions for New, Returning, and Loyal customers; dated purchase history would help track movement between segments.

- **Investigate shipping options:** Express average purchase amount is higher than Standard in this dataset, but the records do not establish that shipping choice causes higher spending.

8. Limitations and Next Steps

Limitations

- No purchase date, cost, profit, units sold, marketing exposure, or explicit online/offline channel field is present.
- 37 review ratings are missing; the notebook does not impute them.
- Customer segment labels follow the supplied SQL logic and may not match standard business definitions.
- Frequency-to-days is an approximate mapping of category labels, not an observed purchase interval.
- Group comparisons are descriptive and do not demonstrate causation.
- The Power BI file has not been visually verified in this report; insert genuine screenshots after checking it in Power BIDesktop.

Recommended next steps

1. Run the notebook from a portable project-relative CSV path and document final row count and null handling.
2. Confirm SQL Server database/table setup and execute all ten queries in SSMS 22; save actual result grids orexported results as evidence.
3. Validate Power BI relationships, measures, slicers, and totals against the CSV/SQL outputs.
4. Add genuine dashboard screenshot evidence.
5. Review the customer-segment business rule and document its rationale.
6. Keep report numbers synchronized with the final executed notebook and SQL results.

9. Tools and Project Artifacts

Artifact / tool	Role in this project
customer_shopping_behavior.csv	Source dataset used for calculations.
customers analysis.ipynb	Python preparation, feature engineering, and SQL Server loading workflow; worked in Jupyter Notebook.
SQLQuery(1).sql	Ten SQL Server queries for descriptive business analysis; run using SSMS 22.

py+sql+tableau_dsbrd(1).pbix	Power BI report file supplied with the project. Open in Power BI Desktop; the filename's “tableau” text does not make it a Tableau workbook.
Python: pandas, NumPy	Data inspection, transformations, and descriptive calculations.
Microsoft SQL Server	Database engine for the prepared customer table and T-SQL analysis.
SQL Server Management Studio (SSMS) 22	SQL client used to connect to SQL Server and execute the query script.
SQLAlchemy, pyodbc, ODBC Driver 17	Python-to-SQL Server connectivity as configured in the notebook.
Microsoft Power BI Desktop	Dashboard/report viewing, validation, and screenshot capture.

Appendix: Key metrics from supplied CSV

Metric	Value
Total records	3,900
Total purchase amount	$233,081
Average purchase amount	$59.76
Median purchase amount	$60.00
Missing review ratings	37
Discount flag = Yes	1,677
Subscription status = Yes	1,053
Unmapped purchase-frequency labels	0
Discount and promo flags identical	Yes, for all supplied records